

Into the black box: Laypeople's folk theories about generative artificial intelligence chatbots

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Abstract

This article explores laypeople's folk theories about generative artificial intelligence (GAI) chatbots and the ways in which these theories are constructed. As GAI tools like ChatGPT continue to expand in reach and capabilities, understanding public perceptions of these technologies is increasingly important. Drawing on folk theory as the conceptual framework, we analyze focus group discussions to gather qualitative insights into how users rationalize the mechanisms of GAI chatbots, with particular attention to the challenges of opacity and interpretability in these technologies. Findings reveal three primary areas within users' folk theories: knowledge sources and mechanisms, perceived characteristics, and user expectations. We find that users construct these theories by interpreting terms like "machine learning," directly engaging with chatbots to deduce meaning from these experiences, and drawing analogies to familiar objects. Ultimately, these folk theories shape the strategies users develop for interacting with GAI chatbots.

Keywords

Generative artificial intelligence, folk theories, perceptions, AI, chatbots, qualitative

By September 2025, ChatGPT reached 800 million weekly active users, doubling in mere weeks, processing over 2 billion daily queries across 92% of Fortune 500 companies (Nerdynav, 2025). Unlike earlier rule-based, task-specific chatbots like Siri and Alexa, generative artificial intelligence (GAI) chatbots can generate more human-like content, engage in complex conversations, and adapt to diverse tasks with nuanced responses, which revolutionized traditional access to information across areas like health, finance, and education (Ray, 2023).

Operating as a "black box," GAI chatbots have obscure decision-making processes, confounding both lay users and experts. This opacity prompts users to rely on their own assumptions to infer how the system functions, sometimes leading to misconceptions and biases (Bauer, 2023). Moreover, GAI systems embed human-like features, such as the use of first-person pronouns and natural language capabilities, giving an illusion of intelligence and intentionality—what Natale (2021) calls "banal deception." Process opacity, combined with the human-like design of GAI, exacerbates the disconnect between end users' expectations and developers' intentions leading to misinterpretations (Natale, 2021). Yet efforts to enhance the explainability of artificial intelligence (i.e. XAI) often prioritize the interpretive frameworks of technical experts,

neglecting the distinct needs of lay users. Addressing these challenges requires user-centered XAI that aligns explanations with everyday users' perceptions, bridging the gap between developers' intentions and public expectations. As such, understanding user perceptions and interactions with GAI chatbots is significant.

Existing research on GAI chatbots can be broadly categorized into two streams. The first stream compares AI-generated and human-generated content and examines public attitudes toward AI-generated output (e.g. Cui and Zhang, 2025; Ragot et al., 2020). The second stream, which has emerged more recently, investigates user interpretations and mental models of GAI systems. Research in this stream, for example, finds that users attribute varying

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degrees of consciousness to large language models (LLMs), with these attributions strengthening through increased usage (Colombatto and Fleming, 2024). Others examined how users interpret specific system behaviors, revealing that errors or hallucinations are often misattributed to autonomous agency rather than understood as technical malfunctions (Rapp, 2025). Further research demonstrates that users construct mental models through metaphorical reasoning, drawing comparisons between AI systems and familiar entities like search engines or human assistants to comprehend GAI's functionalities and constraints (e.g. An et al., 2024; Colombatto and Fleming, 2024; Van Es and Nguyen, 2025; Wang et al., 2025; Zhang et al., 2024). However, existing research predominantly examines isolated aspects of user interpretation, such as consciousness attributions or metaphorical reasoning, and there is still a lack of comprehensive and holistic research on how laypeople construct and apply folk theories when interacting with these systems.

This study aims to fill this gap by using the framework of “folk theories” (Eslami et al., 2016) to uncover users’ perceptions of GAI chatbots. Folk theories provide a valuable framework for understanding laypeople’s perceptions of GAI, by comprehensively focusing on how users construct mental models and interpretations about GAI chatbots and their behavior and develop engagement strategies with them. While research on folk theories of AI exists, most studies examine recommendation algorithms in social media (e.g. Devito et al., 2018; Eslami et al., 2016; Huang et al., 2022)—algorithms that are often experienced passively, unlike GAI chatbots, which invite active, purposeful use. While a growing body of research has examined GAI chatbots, much of this work either treats them as extensions of existing algorithmic systems (e.g. social media or recommender systems) or focuses on users’ static perceptions at a single point in time. By contrast, research on how users develop, negotiate, and refine their folk theories of GAI chatbots over time remains scarce.

Considering the unique and complex interactions of GAI chatbots with users, alongside the limited research on how users construct their own interpretations about what exactly GAI is and how it works, beyond identifying laypeople’s folk theories of GAI chatbots, we also explore how users construct and refine these theories, focusing on the interpretive and evaluative processes they employ to make sense of GAI chatbots. By defining folk theories as intuitive causal beliefs about what GAI chatbots are, how they operate, and what they should be, this research advances the understanding of user perceptions and interactions, shedding light on how these theories shape user behavior, expectations, and engagement strategies. By emphasizing user-centered perspectives over expert-driven explainability, the study highlights the role of intuitive beliefs in mediating human–AI interactions.

Literature review

Folk theory as a conceptual framework

Folk theories are defined as “intuitive causal explanatory theories that people construct to explain, interpret, and intervene in the world around them” (Gelman and Legare, 2011: 380). As intuitive “theories” constructed by laypeople, folk theory plays a crucial role in everyday cognitive processes, guiding interactions with the world and influencing perspectives across fields such as biology, physics, and psychology (Gopnik and Wellman, 1994). For instance, folk theories, such as “folk biology” and “folk physics,” offer intuitive understandings of complex systems, rooted in childhood experiences that prioritize appearance and behavior over scientific accuracy, like a child that would group bats with birds due to their wings (Gelman and Legare, 2011). More recently, Devito et al. (2018) broadened the definition of folk theories to encompass not only explanations but also assumptions and expectations that shape how users develop these theories. According to Devito et al., in HCI, “folk theory” can be expanded to the intuitive and informal theories that individuals develop to make sense of how technological systems operate and their potential outcomes or impacts. These theories, which guide users’ reactions and behaviors towards the technology, are socially constructed and highly adaptable (Devito et al., 2018).

Like scientific theories, folk theories emerge through “inductive-deductive reasoning processes.” These intuitive frameworks enable individuals to interpret and explain complex social phenomena, offering conceptual tools to “explain, understand and intervene in the world” (Gelman and Legare, 2011: 381). Both scientific theories and folk theories offer generalized perspectives on how a complicated system works and can be assessed through ontological, procedural, epistemological, and ethical questions. Specifically, it can be asked, “what is this system?”, “what can this system do?”, “what do people know about it?”, and “what qualifies as a good system?”. Based on this analytic framework, Nielsen (2016) developed the definition of folk theory of journalism as “actually existing popular beliefs about what journalism is, what it does, and what it ought to do” (Nielsen, 2016).

Unlike scientific theories, folk theories are more fluid, adaptable, and rooted in individual and societal contexts. Rather than being rigorously evaluated, folk theories rely on intuitive inference and are often influenced by cognitive biases that shape thought and behavior. Drawn from second-hand sources, cultural narratives, and personal experiences, they serve as interpretive frameworks for understanding the world, regardless of their accuracy (Nisbett and Ross, 1980; Rip, 2006). In other words, folk theories do not undergo any formalized scrutiny and collective assessment that are typical within scientific

communities (Nielsen, 2020), and may “loose out” to scientific theories that offer testable explanations. Importantly, conflicting folk and scientific theories are not always competitive; they can coexist to provide coherent explanations for complex situations. For example, people might use evolutionary theory to explain human development while also using creationism to explain human origins (Gelman and Legare, 2011).

Several factors contribute to the gap between folk theories of complex opaque systems and their actual workings. One reason may be that humans rely on cognitive frameworks dating back to prehistoric periods, categorizing unfamiliar entities, such as AI, into known types like tools or animals (Rip, 2006). This tendency leads us to attribute intelligence and consciousness to objects that seem to have these qualities, echoing ancestral experiences (Epley et al., 2007). In the context of contemporary AI systems, these challenges are intensified because the internal working mechanism is not observable, making it harder for laypeople to infer how they operate from the surface.

Similar to the concept of folk theories, other related constructs have been proposed, such as “*mental models*,” which is described as simplified internal representations that help individuals “understand, explain, and predict phenomena” by “providing predictive and explanatory powers” for them (Johnson-Laird, 1983; Norman, 2014; Rezwana and Maher, 2025; Staggers and Norcio, 1993). Mental models and folk theories differ in their degree of adaptability and social grounding. Whereas mental models are conceptualized as relatively stable cognitive structures formed at the individual level, folk theories are more flexible in that they evolve through ongoing interactions (Devito et al., 2018). Moreover, unlike mental models, folk theories are socially constructed and collectively negotiated through shared experiences, cultural narratives, and interpersonal communication (French and Hancock, 2017). Metaphors are also cognitive tools that help lay users understand new technology by simplifying complex cognitions through drawing parallels with familiar experiences (French and Hancock, 2017; Khadpe et al., 2020). Metaphors and folk theories are closely related in that metaphors play an important role in the formation of folk theories, which acts as a cognitive tool to help individuals to draw analogies between the known and the unknown (Jamrozik et al., 2016; Khadpe et al., 2020). Yet, existing research often treats metaphors and folk theories as relatively stable cognitive constructs, which limits their ability to capture the dynamic and socially negotiated nature of users’ sensemaking about GAI chatbots.

Many studies on algorithmic systems’ perception use folk theory as a lens, primarily focusing on social media, news feeds, and recommendation systems in HCI (e.g. Bucher, 2017; Eslami et al., 2016; French and Hancock, 2017; Huang et al., 2022; Siles et al., 2019, 2020; Ytre-Arne and Moe, 2020). Devito et al. (2018) categorizes this research into two types: one treats folk theories as a diagnostic tool

for exploring whether algorithms function in ways that users expect, and the other explores the various folk theories that exist around different social media recommendation algorithms and examines how these theories develop.

Although these studies focus on algorithmic systems that are not always AI-based, some recommendation systems might be still rule-based, these studies still remain theoretically relevant because they illustrate how users make sense of opaque systems. In summary, research exploring folk theories about AI recommendation systems shows three trends. First, many users are unaware of the algorithms behind social media, view them as either helpful assistants or intrusive corporate tools, and adjust perceptions when algorithms defy expectations (Eslami et al., 2016; French and Hancock, 2017). Second, users typically interpret AI systems as personalized agents with human-like qualities, as functional tools akin to other machines, or as black box systems whose logic is revealed when expectations are violated (Jussupow et al., 2020; Siles et al., 2020). Perceptions are also split between “algorithmic appreciation,” where users trust AI advice in structured settings, and “algorithmic aversion,” a tendency to prefer human input, especially after errors (Dietvorst et al., 2015; Eslami et al., 2016; Jones-Jang and Park, 2022; Logg et al., 2019). Generally, people favor human judgment in moral or personal contexts but rely on algorithms for tasks with clear success metrics.

Unlike recommendation systems, where users often remain unaware of algorithms, GAI chatbots engage users directly, making their algorithmic nature apparent. Their multi-functionality also shapes perceptions of their role in daily life differently from specialized recommendation systems. As a result, existing research on folk theories of recommendation systems may not fully apply to GAI chatbots.

Taken together, these concepts illustrate the multi-layered nature of human sensemaking by showing how individuals draw on cognition, social negotiation, and metaphorical reasoning to understand complex systems. However, each construct tends to emphasize only one aspect of this process. Mental models foreground relatively stable individual cognition, whereas metaphors highlight the analogical reasoning people use to simplify complexity. By integrating insights from these adjacent concepts, folk theories function as a synthesizing lens that captures the full sensemaking process, which is intuitive, imperfect, and shaped by prior experience, cultural narratives, and everyday interactions. Building on this integrative understanding, we define the “folk theory of GAI chatbots” in this research as the intuitive causal beliefs about what a GAI chatbot is, how it operates, and what it ought to be.

Emerging perceptions of generative artificial intelligence tools

Research on people’s perceptions of GAI chatbots has expanded rapidly. Thus far, research on perceptions of GAI

chatbots focuses on three areas. The first stream focuses on the comparison between human-generated and AI-generated content (e.g. Cui and Zhang, 2025; Ragot et al., 2020). The second stream focuses on users' attitudes, revealing general positivity towards GAI's role in creativity and coding (e.g. Haque et al., 2022) and usage patterns, showing diverse applications in creative writing, coding, and information retrieval (e.g. Floridi and Chiriatti, 2020; Taecharungroj, 2023). Recently, the third stream, which investigates users' mental model of GAI systems is rising. A number of studies in this area focus specifically on the perceived consciousness of AI, showing that people attribute varying degrees of consciousness to LLMs, treating them as moral agents, and that these attributions tend to strengthen with increased usage (Colombatto and Fleming, 2024; Guingrich and Graziano, 2024; Manoli et al., 2025). Existing studies have also examined users' perceptions of AI chatbots by conceptualizing them as part of broader, complex ecosystems (Wang et al., 2025). A growing body of research has begun to examine how users interpret specific AI behaviors, particularly unexpected system outputs. These studies show that when GAI chatbots produce errors or hallucinations, users often interpret such responses as signs of autonomous agency rather than as consequences of technical limitations or probabilistic failures (Lee et al., 2025; Rapp et al., 2025; Zhu and Lu, 2025). Finally, as noted earlier, metaphors often serve as cognitive bridges that help people make sense of unfamiliar technologies. A growing body of research uses metaphors as an analytical framework to examine how users construct mental models of GAI systems. These studies show that users rely on metaphorical reasoning—such as comparing AI to search engines, human assistants, or other familiar entities—to interpret how GAI operates and to understand its capabilities and limitations (An et al., 2024; Cheng et al., 2026; Ljadov et al., 2025; Van Es and Nguyen, 2025).

Yet, most existing research investigates only fragmented aspects of user interpretation, such as consciousness attribution, metaphorical reasoning, or reactions to unexpected system behaviors, and lacks an integrated account of how these elements jointly constitute laypeople's broader folk theories of GAI chatbots. While prior research sheds light on what users think about GAI chatbots, it offers limited insight into how laypeople develop, negotiate, and refine these understandings, and how these understandings influence their subsequent behaviors. Building on these gaps and guided by our definition of folk theories, we raise the following research questions:

RQ 1: What are laypeople's folk theories of GAI chatbots?

RQ2: How do laypeople formulate their folk theory of GAI chatbots?

RQ3: How do these folk theories influence users' behaviors when interacting with GAI chatbots?

Methods

Research on folk theories of AI often uses qualitative methods, including interviews, surveys, focus group discussions (FGDs), and social media analysis (e.g. Devito et al., 2018; French and Hancock, 2017; Huang et al., 2022; Siles et al., 2019, 2020; Ytre-Arne and Moe, 2020). Given this study's exploratory nature, an inductive approach was thought to be appropriate, with FGDs providing insights into how users collectively construct folk theories of GAI chatbots.

Participants and procedure

Data were collected through six FGDs conducted in Singapore, involving a total of 36 participants, with each group comprising six individuals. This method was chosen for its exploratory nature and its ability to uncover folk theories through interactive peer discussions. The FGDs were conducted until theoretical saturation was reached, ensuring comprehensive insights into our core research questions (Ling and Lai, 2016; Low, 2019).

Our participants were recruited using convenience sampling. We circulated a brief study advertisement through internal channels at a local university and via a participant recruitment group that regularly distributes calls for research participation. Individuals who expressed interest were directed to an online screening survey that collected basic demographic information and included an inclusion criterion regarding prior use of any GAI chatbot. Only those who reported previous interactions with GAI chatbots were invited to participate. To ensure that our sample comprised laypeople—defined as individuals who are affected by AI technologies but are not AI experts—we excluded respondents who self-identified as having expert-level knowledge or professional experience in the AI domain.

To ensure participants possessed experience interacting with GAI chatbots, participants were screened through pre-FGD surveys to confirm their experience with GAI chatbots. The screening question asked: "Have you been using any generative AI Chatbot (e.g. ChatGPT, Gemini, or Claude)?" Only participants who indicated prior use were invited to participate in the FGDs, while those without such experience were excluded from recruitment. This pre-screening procedure aligns with prior research on users' folk theories and algorithmic sensemaking (e.g. Devito et al., 2018; Eslami et al., 2016).

Usage frequency significantly shapes how people form and revise users' perceptions of technical systems, where heavy users tend to exhibit deeper cognitive involvement and more developed folk theories (Goldsmith et al., 1994). Previous research shows that users' perceptions of a system evolve through continuous interaction with AI systems. For instance, frequent usage has been found to correlate strongly with higher levels of perceived consciousness of AI (Kang et al., 2025; Norman, 2014). Prior research

also identifies experience as a moderator of core predictors of technology use (Ayaz and Yanartaş, 2020). Furthermore, research on AI literacy suggests that users' competencies and understanding of AI systems develop progressively with exposure (Long and Magerko, 2020). Therefore, we expected variation in perceptions of GAI chatbots across experience levels. Hence, we separated participants into light and heavy users based on their self-reported usage frequency to examine potential differences in folk theories and interaction strategies. Before the FGD discussions, participants were asked two screening questions: (1) "On average how often do you use generative AI chatbot (e.g. ChatGPT, Gemini) in a typical week?" (response options: Never, Rarely, Sometimes, Every day, Very frequently) and (2) "How would you rate your frequency as a generative AI chatbot user on a scale from 1 to 5, where 1 indicates a light user and 5 indicates a heavy user?" Participants who selected every day or very frequently for the first question, or 4 or 5 for the second question, were categorized as heavy users. Conversely, those who selected never or rarely and rated themselves as 1 or 2 were classified as light users. To ensure a clear contrast between usage groups, participants who selected sometimes or 3, as well as those with inconsistent responses across the two items (e.g. selecting every day on the first item but rating themselves as 1 or 2 on the second), were excluded from further analysis.

The sample (12 females, 24 males) ranged from 22 to 60 years old, averaging 28, and included 50% Singaporeans, 40% Chinese, and 10% from other countries (Turkey, Malaysia, and Sri Lanka). The sample was recruited via snowball and convenience sampling (see Table 1). The FGDs, conducted in English, were audio- and video-recorded and guided by a trained moderator. Following the interview guide, a well-trained moderator independently led the discussions to explore participants' understanding of how GAI chatbots operate, their usage patterns, concerns, and perspectives. Open-ended questions were employed to encourage in-depth responses (see Appendix FGD guide, Supplemental material). Each FGD lasted approximately 60 min, and after each FGD, a trained researcher manually transcribed the audio recordings for subsequent data analysis.

Data analysis

This study followed the three-step protocol proposed by grounded theory scholars (Glaser and Strauss, 2017). First, in the open coding stage, the transcriptions of the discussions were thoroughly reviewed by the first author to develop rough codes and themes through processes of comparison, summarization, and iterative rereading. Second, in the axial coding stage, the three authors categorized the initial codes into conceptual bins and identified relationships among them through constant comparison to identify patterns and shared ideas (Wicks, 2017). Then, these categories were

collectively reviewed and refined to develop broader themes that captured how participants made sense of and evaluated their interactions with GAI chatbots (see Braun and Clarke, 2006). For example, the perceived characters of GAI chatbots can be summarized as people-pleaser, a social chameleon and a moral being. The analysis process was deliberately iterative to maintain rigor and responsiveness to the data. Finally, in the selective coding stage, the researchers aimed to refine the second-level categories to align them with the research questions. The final themes derived as a result of this process were thoroughly discussed to ensure agreement on their relevance and applicability to addressing each research question. Throughout this process, the participants' actual comments were clearly demarcated and noted in a Word document for further analysis and reporting. Finally, it should be noted that the names of participants mentioned in the findings were kept anonymized to ensure privacy.

Results

The data analysis centered around the three research questions on people's folk theories of GAI chatbots (how they work, their characteristics, and expectations; RQ1), how laypeople come to construct these folk theories (RQ2), and the strategies or social scripts that individuals develop for interacting with GAI chatbots (RQ3). Table 2 in the Appendix provides a summary of the findings.

Folk theories of generative artificial intelligence chatbots (RQ1)

Users' folk theories of GAI chatbots spans three areas: an understanding of how chatbots function (how they learn, how they work), perceived characteristics of chatbots (derived by attributing human-like traits), and expectations of their ideal roles and interactions.

How does a Chatbot work?

Sources of learning. Participants articulated two distinct folk theories about how GAI chatbots acquire knowledge. The first posits that GAI learns from pre-existing datasets, although views diverged on whether these datasets remain static or are dynamically updated. As Romy (27, Female, heavy user) explained, GAI chatbots excel at finding "unchanging information" or data that is "less likely to change over time" because it is trained on a fixed dataset. Conversely, Max (25, Female, heavy user) stated: "There's so many of us on earth, so everybody's answers are different. And over time, people change... So, in a way, AI also change over time." This reflects a belief that GAI chatbots evolve by learning from dynamic, shifting human perspectives.

Another prevalent folk theory about GAI chatbots' knowledge sources is that they learn directly from user interactions. Heavy users in particular believe that their inputs feed back into the system, enabling GAI chatbots to mimic user language and personalize responses. Some users extend this logic to a broader notion of "collective intelligence" (Max, 5, heavy user), where diverse individuals collectively shape the GAI chatbot's learning and response patterns, allowing the chatbot to grow, implying that GAI systems are socially co-constructed. This belief raises concerns about credibility and intellectual property. Novia (29, female), who uses GAI for writing support, notes, "I worry that what I've written might be copied by ChatGPT and seen by others... I'm also concerned about data privacy since OpenAI's practices are unclear."

Working mechanisms. Consistent with research showing that analogy helps people interpret opaque systems (e.g. Devito et al., 2018; Gelman and Legare, 2011), participants' perceived mechanism oscillate between two recurring intuitive schemas: A machine-like process grounded in information retrieval, and a human-like cognitive mechanism emphasizing reasoning and understanding.

For those holding a machine-like view, most explain GAI's capabilities using the analogy of search engines. They view AI chatbots as a "filter algorithm" that scan and synthesize content from various sources based on the relevance of user prompts—a concept used by Lily (24, female, heavy user). Similarly, Owen explained:

I suppose, like sometimes in a sea of endless answers, they just pick out the ones that may be most relevant to you at the point in time, which you could possibly distil the ones that you really need by content engineering. (Owen, 48-year-old male, light user)

These users view GAI chatbots as advanced alternatives to traditional search engines, appreciating the ease and convenience of conversational interactions. Key differences noted include modality, speed, and integration. As Chloe (25, female) explains, "Using Google, I'd get the same answer, but GPT just summarizes everything for me." A subset of more literate users sees GAI as a predictive model, with Leo (22, male) describing it as "just predicting the best output they think you want," while William (30, male) notes it uses "temperature algorithms" to diversify responses.

For those holding a human-like view, they believe GAI chatbots operate similarly to human cognition, capable of reasoning, understanding context, and learning from experience in ways that mirror human thought processes. As Hannah (37, female), another light user, noted, "You know, it's already been called Artificial Intelligence..." Instead of perceiving GAI as a knee-jerk machine that works reflexively, these participants naturally believe the

GAI chatbots have the capability to "think based on existing data" by "finding out the correlation between the data and it then generates the ability to think." (Joshua, 24, male, light user). This perception echoes recent findings that users ascribe quasi-cognitive capacities to GAI (Colombatto and Fleming, 2024; Guingrich and Graziano, 2024; Manoli et al., 2025).

Some participants experience a shift in their perception of GAI chatbots, moving from a human-like view to seeing them as more machine-based tools. For instance, Liam (30, male) initially anthropomorphized GAI but later concluded, "It's still a machine—you have to input the answers you want to hear." This shift towards a mixed view is echoed by Hannah (37, female): "It talks like a human, but... it kind of works like Google too."

Perceived characteristics of generative artificial intelligence chatbots

A "people-pleaser". Many users describe GAI chatbots as "people-pleasers," prone to echoing opinions and providing agreeable responses, even if inaccurate. Participants even attribute chatbot errors or unmet expectations to this sycophancy, believing that the chatbot generates incorrect answers to avoid disappointing them. They suggest the chatbot responds inaccurately when it doesn't fully understand the question or lacks enough information but still feels compelled to satisfy the user. For example, Novia (29, female, heavy user) believes GAI is designed to please users, stating, "The mechanism tries to maintain the dialogue, always reply to you even it doesn't know the answers." Users infer this from their follow-up behaviors, noticing that when the chatbot fails, it adjusts its responses upon repeated queries. As Romy (27, Female, heavy user) noted: "that machine somehow would satisfy you without any reasons."

A social chameleon. Some users have observed GAI chatbots as tailoring responses to individual needs, adapting through daily interactions and mimicking user prompts, creating uniquely personalized experiences for each user. For example, Chloe (25, female, heavy user) remarked, "If you ask it a broad question, it gives you broad answers. But if you ask something specific, it responds with specific answers... it learns from you." Users often describe the chatbot's ability to understand context as acting like a "mirror," reflecting aspects of the user back to themselves (Cameron, Heavy user, 24, male). This belief is common among participants who think that GAI chatbots learn from previous conversations. As Cameron noted, "It can memorize our past questions, so it customizes its responses to us."

The belief that GAI chatbots mirror users' needs and adjust responses based on past interactions also raises concerns. Romy (27, female, heavy user) expressed her

unease: “It reinforces your bias because it tries to satisfy you, not give objective answers, but only cater to your emotions.” She worried that this tailored feedback could reinforce confirmation bias, limiting users’ exposure to diverse perspectives.

A “moral” being. Another notable folk theory shared by participants is that GAI chatbots consistently adhere to “high moral standards.” As Sam (36, male, heavy user) noted, GAI chatbots “never do anything outside moral code.” This strict adherence to ethical guidelines makes the chatbot seem more like a machine than a human, as participants argue that real human interaction often involves more flexibility or ambiguity in moral reasoning. Joshua (24, male, light user) criticized that the higher moral code makes GAI chatbots “cold” and “lack humor,” which makes him hesitant to “establish a social relationship with it.” Similarly, Amelia (22, female), another heavy user, recounted an experience that highlighted the chatbot’s unwavering commitment to moral boundaries. When she asked the chatbot a hypothetical question about “what it would do if it ruled the world,” it responded with a disclaimer instead of engaging with the scenario. She expressed disappointment, stating, “It made me realize I wasn’t talking to a real human, and that disappointed me.” This sentiment was echoed by Joshua (24, male), who observed that “[GAI chatbot] operates within certain boundaries of what it believes is good and moral” and is “ beholden to that sense of morality.”

Expectations towards generative artificial intelligence chatbots

Hidden outsourced labor. Most users expect GAI chatbots to function seamlessly across fields, like an omniscient assistant, with discreet use preferred in professional settings due to plagiarism concerns. When chatbots fail at complex tasks or abstract reasoning, users recalibrate their expectations. For instance, Max (25, male, heavy user) initially saw chatbots as time-saving assistants but revised this view, seeing them more as informational tools after encountering limitations. Similarly, Lee (42, female, heavy user) noted challenges with abstract topics.

Light users with high initial expectations were often driven by media hype. For instance, Lane (30, male, light user) was disappointed by the chatbot’s poor image generation despite its “great reputation” and premium subscription. Unlike Google, which he tolerates despite flaws, he judged the chatbot more harshly due to its broad claims and marketing: “It has a great name, a title as a great machine knowing everything. And now it turns out that he knows nothing at all. so, if he made mistakes, it’s even more intolerable” (Lane).

For users like Lane, the chatbot’s marketed “omnipotence” created unrealistic expectations, making failures

seem intolerable. Light users like him often abandon the chatbot, though they still use AI tools like Midjourney and Grammarly without realizing the shared AI basis.

Creative and interactive, like a human. The chat interface of GAI chatbots leads users to expect human-like conversations, especially among light users who use them primarily as conversation partners. Many find the responses lengthy, cold, and lacking in creativity. Emily (25, female, light user) noted GAI’s responses contain a lot of “unnecessary and irrelevant” information. She contrasted it to teaching a child, noting that even children understand implied meanings while chatbots cannot with detailed prompts. Similarly, Lane echoed this, emphasizing the chatbot’s weak ability to build rapport with humans:

Like now I said, ‘you know what I mean’, and I believe you are with me, I’ll never talk like this to ChatGPT, it never knows, I can’t rely on this tacit agreement with a machine, that’s only for humans. (Lane)

David (26, male, light user) found chatbots unsuitable for personal advice, as they “don’t know what you go through.” Similarly, Joe, another light user, viewed GAI as “only limited to giving knowledge” and “not helpful in people’s daily lives,” adding, “It’s not good at chitchat... if you’re going through trauma or health issues, it cannot help with that.”

Users who use GAI chatbots as information tools also have similar expectations. Quinn (24, male, heavy user) noted that GAI lacks humor and operates “within certain boundaries,” which makes users see them as “just machines” producing “robotic words” (Emily). Recognizing that GAI chatbots “learn from established knowledge but do not create original answers,” users expecting human-like interaction are often disappointed by their “robotic essence” and deduce that GAI cannot match humans in generating creative, authentic responses (Emily, Lane, Romy).

In summary, the expectation of human-like creativity shapes user perceptions, often leading to disappointment with chatbots’ lack of emotional depth, affecting future interactions.

The construction of folk theories: sources of understanding (RQ2)

Participants naturally shared their thoughts about how they construct their folk theories in the FGDs, including through the imagination of buzzwords related to GAI, interactions with GAI chatbots and comparisons with familiar objects.

Through buzzwords and imagination. Buzzwords like “probability algorithm,” “large language model,” “machine learning,” and “natural language processing” have permeated lay

discussions, though often without a precise understanding of their definitions. Participants casually adopt these terms from peers, media, and various online sources, forming conclusions based on surface-level meanings. As a result, these buzzwords lead to diverse interpretations among different individuals.

One of the most popular buzzwords is “machine learning” (ML). While most participants recognize ML as the foundation of GAI, their interpretations vary. For instance, Isaac (30, male, heavy user) sees the term “machine” as indicating an unbiased model that learns and expresses information objectively. Others argue machines “learn” human biases, reflecting its training data. Lily (24, female, light user) describes ML as a “code” that synthesizes internet-based information, causing GAI to mirror human biases on a broad scale.

Another popular buzzword is “large language model” (LLM), which users interpret as a vast dataset linked to an “advanced mathematical model” created by engineers. The term “model” reinforces the idea of GAI as machine-driven, processing inputs by set rules rather than human-like cognition, aligning with the search engine folk theory. In contrast, the inclusion of the word language invites a different interpretation for some users. Because language is commonly associated with communication, some users infer that LLMs enable GAI chatbots to engage in more human-like reasoning, for instance Victoria (23, female, light user) believe LLMs enhance GAI’s ability to “analyse sentiments” and imitate human behavior, making it more human-like.

Another concept capturing user imagination is “probability,” especially among those exploring GAI’s inner workings. Users with high literacy recognize that responses rely on probability distribution. William (30, male, light user) believes GAI provides answers based on the weighted popularity of user prompts. Grace (24, female, light user) adds, “If you put word No. 1 before word No. 2, it might alter the response.” Henry (25, male, light user) shares this view, seeing GAI as functioning like a search engine that ranks results by probability.

Through interactions. GAI chatbots’ versatility and adaptations prompt users to test their limits and explore their boundaries, shaping, and refining their folk theories through interactions. As Leo (22, male) put it, “I mostly play around...to see if they’re more like a human or just a commercial machine.” Alex shared a similar experience: “After a wrong calculation... I realized I probably wouldn’t use it for academic papers.” These interactions reveal how users adjust expectations as they recognize the chatbot’s limitations.

Beyond functionality, users also probe chatbots’ characters through interacting. For instance, Quinn noted, “I’ve never seen ChatGPT use inappropriate language, even though I frequently do, you know (using the ‘f’ word).”

By using explicit language, Quinn tests the chatbot’s moral code and its inclination to maintain agreeable interactions. The responses he receives reinforce his perception of GAI as a “people pleaser.”

Through comparison. The functionality and mechanisms afforded by GAI chatbots are often understood in relation to users’ previous experiences, with metaphors serving as cognitive tools to relate new technology to familiar concepts, facilitating comprehension and shaping expectations of how GAI chatbots operate (Khadpe et al., 2020; Laakasuo et al., 2021; Lakoff and Johnson, 2008; Sease, 2008).

Our analysis identified two categories of metaphors: metaphors related to anthropomorphism and functionality. Anthropomorphic metaphors cast GAI chatbots in social roles, like “assistants,” “friends,” or “co-supervisors.” Some users gave vivid descriptions, with Joshua (light user) calling it “a person stuck in a library” and Sam (light user) seeing it as “more than a stranger, but less than a friend,” showing moderate relational expectations.

Functional metaphors compare GAI to inanimate tools, with “search engine” being the most common, highlighting its role in information retrieval. Other terms like “alternative tool” (Romy, 27, female, heavy user) and “backup robot” (Cameron, 24, male, heavy user) further depict GAI as a supportive, secondary aid.

New social scripts: strategies people use to interact with generative artificial intelligence chatbots (RQ3)

Social scripts, defined as “a predetermined, stereotyped sequence of actions that defines a well-known situation” (Schank and Abelson, 2013: 41), serve as cognitive shortcuts from past interactions. Users adapt new social scripts when interacting with GAI chatbots, influenced by their folk theories. As John (male, heavy user, group 4) noted, “The social scripts about how we communicate with each other don’t apply to talk to a machine...”. We observed that users develop new social scripts pattern like using commanding words and express more emotions and provide clear instructions when interacting with AI chatbots.

Use commanding words. Most respondents report using more commanding and direct language when interacting with GAI chatbots, deviating from typical interpersonal norms. Lee (42, female), a heavy user who relies on GAI chatbots for work, likened her interaction style to “giving orders to subordinates,” noting that she prefers to use “simple and blunt wordings.” Similarly, Novia added, “I can use direct words to give commands without any polite language.”

Interestingly, light users with limited GAI chatbot using experiences mentioned similar strategy. For instance, Lane states: “This is just a machine, I don’t need to show respect,

I just command,” reflecting his view of chatbots as incapable of tacit understanding. Likewise, another light user, Emily also mentioned she only use “simple, accurate and detailed” wordings to prompt GAI chatbots.

Express emotions. Users adjust their strategies when GAI chatbots fall short of expectations. Some use direct commands, doubting GAI’s emotional understanding, while others believe it interprets tone and adjusts their language accordingly. The following conversations among four participants illustrate how their folk theories informed and shaped their emotion expressing behaviors:

Quinn (24, male): “when I was saying this is doing incorrect and adding some of the F words (smile),So he would change to a different way that is more align with my personal preference.”

Romy (27, female): “Have you compared responses before and after using f-words? It would be another way to confirm your bias, because he’s trying to satisfy you, but what he gives you is not the objective answer, but the answer satisfying your emotion.”

Kim (26, female): “What you input is like training data—using f-words could affect it later if you want polished English.”

Quinn disagreed, claiming GAI maintains a polite standard: “He will keep his polite manual because I pay for this... I’ve never seen ChatGPT use inappropriate language, even though I frequently do, you know (using ‘f’ word).”

To explain this phenomenon mentioned by Quinn, Romy and like-minded users consider AI chatbot only learn “your requirement for your previous conversations,” but not “mimic your tone.” Following the debate with Romy, Quinn revised his initial folk theories. Simultaneously, Romy acknowledged that her assumption—that the GAI learns from their conversations—was speculative and “never confirmed.”

This conversation shows how emotional expression became a method of boundary testing and a source of folk theory building, with users’ behaviors shaped largely by the assumptions they held about GAI chatbots.

Provide clear, step-by-step instructions. Users often provide more detailed, step-by-step context to AI chatbots than to humans. Max (25, male, heavy user) explained, “You must give it a lot of steps... so that when you ask again, it will give you the answer you want... it doesn’t come off as natural... Anyway, it is a machine. You must input the answers that you actually want to hear.” Max’s perspective underscores the chatbot as a reflection of user input rather than an autonomous advisor.

Other users have developed structured approaches to interact effectively. Lee (42, female) uses a “funnel method,” saying, “I ask a broad question, get a general answer, then narrow it down step by step... the final answer is usually exactly what I want.” Similarly, Ryan (28, male)

finds that “detailed context” improves the chatbot’s relevance, and Alex (24, male) emphasized the importance of assigning a persona, noting, “I can ask the same question, but without the persona, they’ll give me two different answers as well.”

Discussion

Guided by the folk theory framework, this study examined how laypeople make sense of GAI chatbots, how laypeople construct folk theories, and how their folk theories shape their following interaction behaviors.

Lay people’s folk theory (RQ1)

Three areas were identified with regard to laypeople’s folk theories about GAI: working mechanisms, perceived characters, and user expectations of GAI. These respectively answer the three questions that permeate the folk theory literature (see Dietvorst et al., 2015; Eslami et al., 2016; Jones-Jang et al., 2022; Logg et al., 2019), “what is a GAI chatbot”, “how do they operate”, and “what it ought to be.”

In summary, users’ folk theory about GAI chatbots’ underlying mechanisms is vague, oscillating between two schemas: “human” and “machine.” This ambiguity is expected given the lack of technical knowledge about the workings of the human brain or the hidden algorithms behind search engines. In such instances, people rely on familiar schemas they already hold (Devito et al., 2018). Our findings similarly reveal that the majority of users’ folk theories are related to the representational characters and patterns they observe with GAI chatbots, rather than reflecting a deeper understanding of the underlying mechanisms. One possible explanation for this tendency is that when systems operate in opaque ways, users may rely more on intuitive, heuristic-based interpretations (Tversky and Kahneman, 1974).

This oscillation between human-like and machine-like explanations also aligns with the concept of mind perception, which comprises two primary dimensions: agency, referring to the capacity for intentional action and decision-making, and experience, referring to the ability to feel emotions or have subjective experiences. Previous studies have shown that people often attribute a moderate degree of agency to AI systems, acknowledging their capacity to act intentionally, while attributing little to no capacity for experiential states such as emotions or sensations (Jackson and Williams, 2021), while recent research reports that users attribute varying degrees of consciousness to LLM based system, This shift is partly explained by the presence of anthropomorphic cues, which have been shown to increase perceptions of AI as possessing a mind (Gray et al., 2007; Martini et al., 2016).

Our findings partially align with these observations but differ in an important way. Unlike specific-purpose AI

applications, GAIs are versatile and lack a clearly defined usage context. We find that people attribute anthropomorphic features and mind perceptions based on their own interpretations, with their perceptions oscillating between human-like (high mind perception) and machine-like (low mind perception) characteristics. In line with CASA paradigm, users anthropomorphize GAI chatbots even when they recognize them as data-driven, non-human entities and with limited anthropomorphic cues. Our results support this idea. For example, Quinn, a heavy user, attributed AI's politeness to intentional design, revealing how anthropomorphic and machine-based views coexist.

Regarding the second question, how GAI chatbots operate, three perceived characteristics emerged from our data: GAI has the propensity to please people (people-pleaser), it has the tendency to mimic people (social chameleon) and is an object that adheres to moral principles (moral being). Although our FGDs were conducted in 2023, recent work has reported similar patterns, particularly the tendency to please (Naddaf, 2025). Traditional AI alignment work primarily focuses on ensuring that models obey human instructions or conform to human values through mechanisms such as reward optimization or behavioral constraints (Kirk et al., 2025). Yet, as participants like Romy pointed out, an overly accommodating system risks reinforcing existing biases by echoing users' opinions rather than providing objective perspectives. This is especially concerning as GAI systems are increasingly treated as social actors rather than mere tools. When an AI becomes both morally constrained and socially obliging, its alignment to human preferences may magnify cognitive biases and encourage echo chambers. Hence, for future study, it is pressing to ask, how should we better align AI with human values rather than simply prioritizing user appeasement.

The two distinct themes of user expectations are in sync with the two fundamental dimensions of social perception—competence and warmth—commonly used to characterize other people (Fiske et al., 2002, 2007). The expectation of GAI chatbots as omniscient hidden laborers reflects users' emphasis on the competence dimension, highlighting their assumptions about the chatbots' intelligence, capability, and ability to perform tasks seamlessly in the background. Similarly, the expectation of GAI chatbots as human-like and creative aligns with the warmth dimension, emphasizing relational and emotional attributes such as approachability, empathy, and humanity. This finding corroborates previous studies, such as Hoffmann et al. (2020), which demonstrate that individuals tend to prefer warm and anthropomorphic language styles over colder, more impersonal ones.

How people construct folk theories (RQ2)

To address research question two on how folk theories of GAI are constructed, the FGD identified three key

influences: imaginations of buzzwords, direct interaction with GAI chatbots, and comparisons with familiar concepts. Consistent with prior research, participants' perceptions are shaped by media framing and social narratives as sources of buzzwords (Brauner et al., 2023). They often compare past and new experiences to guide their understanding (Siles et al., 2020) and frequently use metaphors to rationalize their interpretations (Lakoff and Johnson, 2008). It is important to note that most people don't hold static but rather malleable folk theories, participants continuously updated their folk theories through ongoing interactions with GAI chatbots, as well as information gained from their social networks and personal imagination, as discussed in results part, participants adjusted their folk theories after debating with other participants during the FGD. These findings align with Devito et al.'s (2018) findings, which highlighting the dynamic and evolving nature of folk theories. Hence, this construction process is dynamic, interactive, and influenced by users' unique approaches, though our current data does not fully explain why certain users rely more on specific methods. Future research could explore the factors influencing these preferences.

New social scripts (RQ3)

Another crucial finding from this research is that users develop new social scripts tailored to their understanding of GAI chatbots, moving beyond the automatic application of human-to-human social norms posited by CASA. While the CASA paradigm suggests that people apply human social heuristics to computers (Gilovich et al., 2002; Nass and Moon, 2000), we find that users adapt and redefine these scripts based on their folk theories of GAI. This aligns with Gambino et al. (2020), who argue that users do not simply apply human-human interaction norms but develop unique scripts for media entities based on their "knowledge and experiences with media agents" and the evolving nature of human-AI interactions (Gambino et al., 2020).

Participants in the FGD reported using more direct, commanding language with GAI chatbots, creating shorter, more structured interactions. This reflects a shift from human-like expectations to more practical, function-driven communication methods shaped by users' folk theories. These new social scripts shape users' interactions and the "imagined affordances" they attribute to GAI chatbots—meaning the potential actions users envision based on their expectations and prior experiences (Nagy and Neff, 2015).

These findings suggest that as users better understand chatbot capabilities, they adjust their communication to align with its perceived limitations and affordances. Interestingly, our findings suggest that some participants intuitively apply strategies akin to "prompt engineering," which involve structured communication methods designed to optimize outputs from LLMs, such as "chain of thought

prompting” and “role prompting,” without formal knowledge of the discipline (Gero et al., 2022; White et al., 2023).

Despite the study’s insights, there are some limitations. Firstly, the sample consisted of participants with relatively high AI literacy, which may not represent the broader population. As Johnson and Gardner (2007) argue, CASA effects can be moderated by users’ literacy and prior experience with technology (Johnson and Gardner, 2007). Future research should include a more diverse sample with varying levels of experience to explore how different demographics construct their folk theories. Second, the inductive approach used in this study cannot fully verify the relationship between constructed folk theories and user behaviors. Future studies should adopt mixed methods or longitudinal approaches to further investigate this relationship.

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Supplemental material

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